Exercise Sheet

Lecture 12

Tensors and the Tensor Product

The core uses finite-dimensional algebraic tensor products. $\otimes$ is the tensor (Kronecker) product; $|ij\rangle = |i\rangle \otimes |j\rangle$. Quantum kets in this sheet are nonzero pure-state representatives, so non-factorization tests pure-state entanglement only. General Hilbert composites require completion, and mixed-state separability is a convex-decomposition condition. A $\star$ marks a harder problem.

Core tutorial route

Work **12.1, 12.18, 12.5, 12.8, 12.11, 12.12, 12.19, and 12.23** in that order (about 85–100 minutes before discussion). This route covers an elementary tensor, an operator Kronecker calculation, a trace proof, an authentic universal-property factorization, real tensor type/metric classification, contraction invariance, product-state factorization, and a coefficient-rank nonfactorization proof.

Additional practice: 12.2–12.4, 12.6–12.7, 12.9–12.10, 12.14, 12.22, 12.24, 12.26, and 12.28. **Bridge:** 12.13 and 12.15–12.17 (symmetric/exterior tensors and determinants). **Extension:** 12.20–12.21, 12.25, and 12.27 (singlet symmetry, partial trace/reduced states, and Schmidt decomposition). Bridge and extension tasks do not count toward core progress.

Tensor and Kronecker products

Exercise 12.1 (computational) Compute the Kronecker product $v \otimes w$ for $v = (1, -1)$ and $w = (2, 3)$, and write the two-qubit state $|1\rangle \otimes |0\rangle$ as a column.

Exercise 12.2 (computational) Form $\sigma_z \otimes \sigma_x$ as a 4×4 matrix.

Exercise 12.3 (computational) Compute the Kronecker product $v \otimes w$ in $\mathbb{R}^2 \otimes \mathbb{R}^2 \cong \mathbb{R}^4$ for $v = (2, 1)$ and $w = (3, -1)$.

Exercise 12.4 (computational) Compute $v \otimes w$ in $\mathbb{C}^2$ for $v = (1, i)$ and $w = (2, -i)$.

Exercise 12.5 (computational) Form $A \otimes \sigma_z$ as a 4×4 matrix, where $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

Exercise 12.6 (computational) A matrix A is 3×3 with $\operatorname{tr} A = 2$ and $\det A = 4$; a matrix B is 2×2 with $\operatorname{tr} B = -1$ and $\det B = 3$. Compute $\operatorname{tr}(A \otimes B)$ and $\det(A \otimes B)$.

Exercise 12.7 (computational) Find the eigenvalues of $A \otimes B$, where $A = \operatorname{diag}(2, 3)$ and $B = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Exercise 12.8 (proof) Prove that $\operatorname{tr}(A \otimes B) = (\operatorname{tr} A)(\operatorname{tr} B)$.

Exercise 12.9 ($\star$) Show that if $Av = \lambda v$ and $Bw = \mu w$, then $v \otimes w$ is an eigenvector of $A \otimes B$ with eigenvalue $\lambda\mu$. Deduce the eigenvalues of $\sigma_z \otimes \sigma_z$.

Exercise 12.10 (proof) Show that if A and B are invertible then so is $A \otimes B$, with $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Tensors as multilinear objects

Exercise 12.11 (conceptual) State the type (p, q) of each: a vector, a covector, the metric, a linear operator, and a trilinear form. Interpret "metric" here in the real bilinear setting; state separately why a complex Hermitian inner product is not an ordinary bilinear $(0, 2)$ tensor.

Exercise 12.12 (proof) Show that the contraction $T^i{}_i$ of a $(1, 1)$ tensor equals the trace and is basis-independent.

Exercise 12.13 (computational) Split $e_1 \otimes e_2$ in $V \otimes V$ into its symmetric and antisymmetric parts.

Exercise 12.14 (computational) The operator $T = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$ acts on a 2-dimensional space. Write its $(1, 1)$ components $T^i{}_j$ and compute the contraction $T^i{}_i$.

Exercise 12.15 (computational) For $u = (2, 1)$ and $v = (3, 4)$ in $\mathbb{R}^2$, compute the wedge $u \wedge v$ in $\Lambda^2\mathbb{R}^2$.

Exercise 12.16 (computational) Split the tensor $\tau \in V \otimes V$ with component grid $\left(\begin{smallmatrix} 1 & 4 \\ 2 & 3 \end{smallmatrix}\right)$ into its symmetric and antisymmetric parts (give the grid of each).

Exercise 12.17 (computational) For $\dim V = 4$, compute $\dim S^2V$ and $\dim \Lambda^2 V$, and check they sum to $\dim(V \otimes V)$.

Exercise 12.18 (proof) Let $B\colon \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$ be $B((x_1, x_2), (y_1, y_2)) = (x_1y_1,\ x_1y_2 + x_2y_1)$. Construct the unique linear map $\widetilde{B}\colon \mathbb{R}^2 \otimes \mathbb{R}^2 \to \mathbb{R}^2$ supplied by the universal property: give its values on $e_i \otimes e_j$, then verify $\widetilde{B}(v \otimes w) = B(v, w)$ for arbitrary v, w and explain uniqueness.

Entanglement

Exercise 12.19 (computational) Decide whether $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ is a product state; if so, factor it.

Exercise 12.20 ($\star$) Prove that $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ (the singlet) is entangled.

Exercise 12.21 (computational) For the Bell state $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, compute the reduced density matrix of the first qubit.

Exercise 12.22 (computational) Decide whether $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$ is a product state; if so, factor it.

Exercise 12.23 (computational) Decide whether $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ is entangled.

Exercise 12.24 (computational) Compute $(\sigma_x \otimes \sigma_z)|10\rangle$.

Exercise 12.25 (computational) Compute the reduced density matrix ρ_A of the first qubit of the product state $|+\rangle \otimes |0\rangle$.

Exercise 12.26 (computational) For the Bell state $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, compute the expectation $\langle\Phi|\, \sigma_z \otimes \sigma_z \,|\Phi\rangle$.

Exercise 12.27 ($\star$) *(Schmidt decomposition.)* Let $|\psi\rangle = \frac{1}{\sqrt{10}}\big(|00\rangle + 2|01\rangle + 2|10\rangle + |11\rangle\big)$.

1. Show from its coefficient grid that $|\psi\rangle$ is entangled.

2. Diagonalize the grid to write $|\psi\rangle = \lambda_1|a_1\rangle \otimes |b_1\rangle + \lambda_2|a_2\rangle \otimes |b_2\rangle$ with orthonormal $|a_k\rangle$, $|b_k\rangle$ and $\lambda_k > 0$ (its *Schmidt decomposition*).
3. Verify that the reduced state of the first qubit has eigenvalues λ_k^2.

Exercise 12.28 (proof) Prove that a two-qubit state $|\psi\rangle = \sum_{ij} M_{ij}|ij\rangle$ is a product state if and only if its coefficient grid M has rank at most one.